

```

import tensorflow as tf
import numpy as np
from tensorflow.keras.layers import Input, Conv2D, LeakyReLU, Add, AveragePooling2D, MaxPooling2D
from tensorflow.keras.models import Model

def build_and_run_model():
    input_data = np.array([
        [1, 2, 3, 0, 1],
        [0, 1, 2, 3, 0],
        [3, 0, 1, 2, 1],
        [1, 2, 3, 0, 0],
        [0, 1, 0, 1, 2]
    ], dtype=np.float32).reshape(1, 5, 5, 1)

    print("Input Data Shape:", input_data.shape)
    print("Input Data Matrix:\n", input_data[0, :, :, 0])
    print("-" * 50)

    input_tensor = Input(shape=(5, 5, 1))

    x = Conv2D(filters=1, kernel_size=(3, 3), padding='same',
               kernel_initializer='ones', use_bias=False, name="Initial_Conv")(input_tensor)
    x = LeakyReLU(name="Initial_LeakyReLU")(x)

    path_a = x

    path_b = Conv2D(filters=1, kernel_size=(3, 3), dilation_rate=3, padding='same',
                     kernel_initializer='ones', use_bias=False, name="Dilated_Conv_r3")(x)
    path_b = LeakyReLU(name="LeakyReLU_r3")(path_b)

    path_b = Conv2D(filters=1, kernel_size=(3, 3), dilation_rate=2, padding='same',
                     kernel_initializer='ones', use_bias=False, name="Dilated_Conv_r2")(path_b)
    path_b = LeakyReLU(name="LeakyReLU_r2")(path_b)

    path_b = Conv2D(filters=1, kernel_size=(3, 3), dilation_rate=1, padding='same',
                     kernel_initializer='ones', use_bias=False, name="Dilated_Conv_r1")(path_b)
    path_b = LeakyReLU(name="LeakyReLU_r1")(path_b)

    merged = Add(name="Merge_Add")([path_a, path_b])

    intermediate = Conv2D(filters=1, kernel_size=(3, 3), padding='same',
                           kernel_initializer='ones', use_bias=False, name="Intermediate_Conv")(merged)

    pool_avg = AveragePooling2D(pool_size=(2, 2), strides=(2, 2), padding='valid', name="Avg_Pool")(intermediate)
    pool_max = MaxPooling2D(pool_size=(2, 2), strides=(2, 2), padding='valid', name="Max_Pool")(intermediate)

    final_output = Add(name="Final_Output")([pool_avg, pool_max])

    model = Model(inputs=input_tensor, outputs=final_output)

    print("\nModel Summary:")
    model.summary()

    prediction = model.predict(input_data)

    print("\n" + "-" * 50)
    print("FINAL OUTPUT FEATURE MAP VALUES")
    print("-" * 50)
    print(prediction[0, :, :, 0])

if __name__ == "__main__":
    build_and_run_model()

```

Input Data Shape: (1, 5, 5, 1)

Input Data Matrix:

```
[[1. 2. 3. 0. 1.]
 [0. 1. 2. 3. 0.]
 [3. 0. 1. 2. 1.]
 [1. 2. 3. 0. 0.]
 [0. 1. 0. 1. 2.]]
```

Model Summary:

Model: "functional_1"

Layer (type)	Output Shape	Param #	Connected to
input_layer_1 (InputLayer)	(None, 5, 5, 1)	0	–
Initial_Conv (Conv2D)	(None, 5, 5, 1)	9	input_layer_1[0]...
Initial_LeakyReLU (LeakyReLU)	(None, 5, 5, 1)	0	Initial_Conv[0][...]...
Dilated_Conv_r3 (Conv2D)	(None, 5, 5, 1)	9	Initial_LeakyReLU...
LeakyReLU_r3 (LeakyReLU)	(None, 5, 5, 1)	0	Dilated_Conv_r3[0]...
Dilated_Conv_r2 (Conv2D)	(None, 5, 5, 1)	9	LeakyReLU_r3[0][...]...
LeakyReLU_r2 (LeakyReLU)	(None, 5, 5, 1)	0	Dilated_Conv_r2[0]...
Dilated_Conv_r1 (Conv2D)	(None, 5, 5, 1)	9	LeakyReLU_r2[0][...]...
LeakyReLU_r1 (LeakyReLU)	(None, 5, 5, 1)	0	Dilated_Conv_r1[0]...
Merge_Add (Add)	(None, 5, 5, 1)	0	Initial_LeakyReLU... LeakyReLU_r1[0][...]...
Intermediate_Conv (Conv2D)	(None, 5, 5, 1)	9	Merge_Add[0][0]
Avg_Pool (AveragePooling2D)	(None, 2, 2, 1)	0	Intermediate_Conv...
Max_Pool (MaxPooling2D)	(None, 2, 2, 1)	0	Intermediate_Conv...
Final_Output (Add)	(None, 2, 2, 1)	0	Avg_Pool[0][0], Max_Pool[0][0]

Total params: 45 (180.00 B)

Trainable params: 45 (180.00 B)

Non-trainable params: 0 (0.00 B)

1/1 ————— 0s 66ms/step

FINAL OUTPUT FEATURE MAP VALUES

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[[13836.5 17082.75]
 [17054.25 21165.  ]]
```